1. We can count the integer solutions to

$$x_1 + x_2 + x_3 + x_4 = 8, \qquad x_i \geq 1,$$

with $x_4$ odd. Write $x_4 = 2k+1$ with $k \geq 0$. Then we find

$$x_1 + x_2 + x_3 = 7 - 2k, \qquad x_1, x_2, x_3 \geq 1.$$

We need $7 - 2k \geq 3 \Rightarrow k = 0, 1, 2.$

$$\begin{aligned} k = 0 : x_4 = 1,\ x_1 + x_2 + x_3 = 7 \ &\Rightarrow \binom{6}{2} = 15, \\ k = 1 : x_4 = 3,\ x_1 + x_2 + x_3 = 5 \ &\Rightarrow \binom{4}{2} = 6, \\ k = 2 : x_4 = 5,\ x_1 + x_2 + x_3 = 3 \ &\Rightarrow \binom{2}{2} = 1. \end{aligned}$$

Therefore, through addition of previous, we have the total number of integer solutions:

$$15 + 6 + 1 = 22.$$

2. In $(2x - 3y + 4z)^{19}$, the coefficient of $x^9 y^3 z^7$ is the multinomial coefficient (quite trivially)

$$\frac{19!}{9!\,3!\,7!}(2)^9(-3)^3(4)^7.$$

Thus the coefficient follows the format of the theorem (where we pull the negative from the $-3$):

$$-\frac{19!}{9!\,3!\,7!}\,2^9\,3^3\,4^7$$

3. Trivially, we find the letters in `MISSISSIPPI` are: $M^1, I^4, S^4, P^2$ (total 11 letters). We wish to arrange the non-$I$ letters: $M, S, S, S, S, P, P$ (7 letters), giving

$$\frac{7!}{4!\,2!} = 105$$

distinct arrangements. We find these 7 letters create 8 slots (including ends) in which we can place $I$'s:

$$\_\,L\,\_\,L\,\_\,\cdots\,L\,\_$$

To avoid consecutive $I$'s, choose 4 of these 8 slots and place one $I$ in each:

$$\binom{8}{4} = 70.$$

Therefore, we find the number of valid permutations is

$$105 \cdot 70 = 7350.$$

4. From the image, we find Leon will start at $S = (0,2)$, the store is at $C = (4,4)$, and the friend's house is at $T = (8,8)$. A shortest lattice path uses only steps $E = (1,0)$ and $N = (0,1)$.

Because Leon must pass through $C$, the number of valid shortest paths is

$$\#(S \to C \to T) = \#(S \to C) \cdot \#(C \to T),$$

counting only paths that do not cross the river.

We begin by calculating from $S = (0,2)$ to $C = (4,4)$: This requires 4 east steps and 2 north steps, so the total number of shortest paths is

$$\binom{6}{2} = 15.$$

Along this segment, we find the only blocked river point that is reachable is $(4,2)$ (points $(3,0), (3,1)$ are below $y = 2$ and cannot be reached). Therefore, the number of shortest paths that go through $(4,2)$ is

$$\#(S \to (4,2)) \cdot \#((4,2) \to C) = 1 \cdot 1 = 1,$$

since $S \to (4,2)$ must be $EEEE$ and $(4,2) \to (4,4)$ must be $NN$. Hence

$$\#(S \to C) = 15 - 1 = 14.$$

We will now calculate $C = (4,4)$ to $T = (8,8)$ avoiding the river (reflection principle): This segment requires 4 east steps and 5 north steps. The blocked river points that are reachable on this segment are

$$(5,4), (6,5), (7,6), (8,7),$$

which are exactly the lattice points on the line $y = x - 1$ inside the rectangle from $C$ to $T$. Therefore, avoiding these points is equivalent to staying strictly above the line $y = x - 1$.

We will shift coordinates by $(x', y') = (x - 4, y - 4)$. Then, we count monotone paths from $(0,0)$ to $(4,5)$ that never go below the diagonal $y' = x'$. Total monotone paths:

$$\binom{8}{4} = 70.$$

By the reflection principle, paths that ever go below $y' = x'$ are in bijection with monotone paths from $(0,0)$ to the reflected endpoint $(5,4)$, hence there are

$$\binom{8}{5} = \binom{8}{4-1} = 56$$

bad paths. Therefore the good paths are

$$\#(C \to T) = 70 - 56 = 14.$$

Finally, we will multiply the two independent segments:

$$\#(S \to C \to T) = 14 \cdot 14 = 196.$$